Sultan Mahamud Opu

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Professional Profile

Computer Science and Engineering graduate with research interests in Machine Learning, Explainable Artificial Intelligence (XAI), and responsible AI. Experienced in collaborative research, model evaluation, and software development, with contributions to IEEE conference research and dataset development. Seeking an academic role where I can contribute to effective teaching, student mentoring, and impactful research.

Education

- **Hajee Mohammad Danesh Science and Technology University** — *March, 2022 – September 2026*
  B.Sc. in Computer Science and Engineering — Dinajpur, Bangladesh
  - CGPA: 3.694/4.00; ranked 5th in the batch
- **Dr. Mahbubur Rahman Mollah College** — *2018 – 2020*
  Higher Secondary Certificate (Science) — Dhaka, Bangladesh
  - GPA: 4.67/5.00
- **Bandura Holy Cross High School & College** — *2016 – 2018*
  Secondary School Certificate (Science) — Dhaka, Bangladesh
  - GPA: 4.89/5.00

Experience

- **AU-Ltd** — *March 2026 – Present*
  Mobile Application Developer
  - Built FitWell-Flow from initial development through Google Play and Apple App Store release, implementing authentication, profile management, REST API integration, activity tracking, clock-in/out, and leave management.
  - Implemented user authentication features for the Eshogol mobile application.

Publications & Research Contributions

Publications: **C=Conference, U=Under Review, E=Extended Abstract, D=Dataset Contribution**

[U.1] Najma Akter Lopa, **Sultan Mahamud Opu** and Others, *"Towards Responsible Clinical AI: An XAI-Driven Framework for Cardiovascular Disease Risk Assessment with Clinical Decision Support",* 29th International Conference on Computer and Information Technology (ICCIT 2026), IEEE

[C.1] Sourav Roy, Najma Akter Lopa, **Sultan Mahamud Opu** and Others, *"Explainable Machine Learning Framework for Early Detection of Depression in University Students",* 2026 IEEE International Conference on Engineering and Frontier Technologies (ICEFronT 2026), IEEE (Accepted)

[D.1] Najma Akter Lopa, **Sultan Mahamud Opu**, and Others (2026), *"A Cross-Sectional Dataset of Sleep Disorder Indicators Among Bangladeshi University Students"*, **[Thesis-based Primary Dataset]**, *Mendeley Data*, V1. DOI: 10.17632/9djgjvtj8d.1.

[C.2] Najma Akter Lopa, **Sultan Mahamud Opu**, and Others (2026). *"AI-Driven Analysis of Sleep Disorders in Bangladeshi University Students Using Clinical and Behavioral Insights"*. In 5th IEEE International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2026), **[Thesis-based Conference]**, DOI: 10.1109/ICECTE69292.2026.11429299

[E.1] Sourav Roy, Najma Akter Lopa, **Sultan Mahamud Opu**, and Others, *"Enhancing Early Detection of Depression in Academic Settings: An XAI-Guided Machine Learning Approach for Bangladeshi University Students"*. In Scholars' Summit 2025, HSTU Research Society. **[Accepted]**, Certificate

[E.2] Najma Akter Lopa, **Sultan Mahamud Opu**, and Others,*"Towards an AI-Driven Framework for Effective Sleep Disorder Prediction Analyzing Clinical and Behavioral Insights",* RUEC 1st International Research Conference (2025). **[Accepted]**, Certificate

Thesis: **D=Defense Completed, W=Working on it**

[D.1] **Sultan Mahamud Opu**, and Others (2026). *"Sleep Disorder Risk Assessment and AI-Based Clinical Decision Support Among University Students"*. In **Level 4 Semester II** of B.Sc. (Engineering) in Computer Science and Engineering, supervised by Pankaj Bhowmik and Dr. Masud Ibn Afjal.

[D.2] **Sultan Mahamud Opu**, and Others (2025). *"A Comparative Analysis of Gender Disparity in Programming Skills Using Educational Data Mining"*. In **Level 4 Semester I** of B.Sc. (Engineering) in Computer Science and Engineering, supervised by Pankaj Bhowmik and Dr. Masud Ibn Afjal.

PROJECTS

• **StudyForge AI (Hackathon Win)** *2026*
Tools: Python, LangChain, Gemma, ElevenLabs, Flask Live Demo

- Built an AI-powered learning platform featuring an AI tutor, study planner, smart quizzes, and a PDF reader.
- Implemented personalized study plans, instant-scored quizzes, and PDF-grounded chatbot responses.
- Added Bangla-English support, text-to-speech, and PDF downloads.

• **HSTU Student Complaint Management System** *June 2026 – August 2026*
Tools: HTML, CSS, JavaScript, Backend, MySQL Live Demo

- Built a web-based complaint management system with unique complaint IDs and end-to-end tracking.
- Implemented complaint-status updates and student feedback.
- Developed an admin panel for complaint review, status management, and monitoring.

• **Personal Local Cloud Storage and File Sharing System** *May 2026 – July 2026*
Tools: Python, FastAPI, MySQL, HTML, CSS, JavaScript, Ngrok Personal Cloud

- Developed a local cloud platform for file storage, organization, and sharing.
- Implemented QR/IP-based access, folder management, dashboards, and file recovery.
- Added encrypted storage and external sharing through Ngrok.

• **Diabetes Prediction System** *January 2026*
Tools: Python, Scikit-learn, SHAP, Gradio Live Demo

- Developed a diabetes prediction model using 2,019 records and 15 features, achieving 95.67% accuracy.
- Applied preprocessing, scaling, normalization, and Bayesian hyperparameter optimization.
- Evaluated performance using ROC-AUC, stratified 5-fold cross-validation, and stress testing; interpreted feature importance with SHAP.

• **Tic-Tac-Toe AI Game** *May 2025*
Tools: HTML, CSS, JavaScript, Minimax Live Demo

- Built an interactive Tic-Tac-Toe game with three AI difficulty levels.
- Implemented the Minimax algorithm with alpha-beta pruning and deployed the game through GitHub Pages.

• **HSTU_S&T_Hub: Real-Time Communication System (Project Showcasing Win)** *January 2025 – June 2025*
Tools: Flutter, Android, IOS, Web, Supabase, Firebase HSTU_S&T_Hub

- Built a platform for teachers and class representatives to coordinate schedules, room allocations, and academic communication.
- Integrated real-time messaging, role-based access control, and an AI chatbot using Firebase and Supabase.

LEADERSHIP & PROFESSIONAL ACTIVITIES

• **IEEE Computer Society HSTU Student Branch Chapter** *March 2026 – September 2026*
Treasurer

• **IEEE Computer Society HSTU Student Branch Chapter** *January 2025 – March 2026*
Publication Secretary

• **Programmers Arena (HSTU)** *September 2025 – September 2026*
Office Secretary

• **Programmers Arena (HSTU)** *January 2025 – September 2025*
Publication and IT Secretary

• **Programmers Arena (HSTU)** *February 2024 – January 2025*
Executive Committee Member

- **HSTU Research Society (HSTURS)** *2023 – 2024*
  Executive Committee Member
- **CSE Club of HSTU** *January 2022 – January, 2023*
  IT Secretary
- **Team Matrix – Ethical Hackers** *January 2022 – 2026*
  Campus Ambassador

HONORS & AWARDS

- **Honorable Dean's Award** *2026*
  - Awarded by the Faculty of Computer Science and Engineering, HSTU.
- **Winner – Hacktoberfest Hack Day Dinajpur 2026** *October 2026*
  - Best Open-Source AI Project.
  - Best Project Built with ElevenLabs.
- **Runner-up – Scholars' Summit 2025** *November 2025*
  - XAI-based depression detection project, Segment A (Science & Engineering), HSTU Research Society.
- **4th Place – Rangpur Divisional Project Showcasing** *June 2025*
  - HSTU_T&S_HUB: A Real-Time Communication System.
- **Participant – Learnathon, Brain Station 23** *January 2025*
- **7th Place – Rangpur Divisional Collegiate Programming Contest (BRUR)** *October 2024*

SKILLS

- **Languages:** Strong reading, writing, and speaking competencies in Bangla and English.
- **Programming Languages:** C, C++, Java, Python, Dart.
- **Web Technologies:** HTML5, CSS3, FastAPI.
- **Frameworks:** Flutter, JavaFX, Streamlit.
- **Database Systems:** MySQL, Firebase, Supabase.
- **Data Science & Machine Learning:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, SHAP, LIME.
- **Machine Learning:** Classification, Regression, Ensemble Learning, Feature Engineering, Feature Selection, Hyperparameter Optimization, Cross-Validation.
- **Specialized Area:** Explainable AI (XAI), Stacking Ensemble, Bayesian Optimization, Statistical Analysis, ML Model Evaluation.
- **Deep Learning & NLP:** Neural Networks, CNN, RNN, LSTM, Transformers, Natural Language Processing, Text Classification, Text Preprocessing.

REFERENCES

1. **Pankaj Bhowmik**
   Assistant Professor, Department of Computer Science and Engineering
   Hajee Mohammad Danesh Science & Technology University, Dinajpur
   Email: pankaj.cshstu@gmail.com
   Phone: +8801791848439
   Relationship: Thesis supervisor and research collaborator

2. **Dr. Masud Ibn Afjal**
   Professor, Department of Computer Science and Engineering
   Hajee Mohammad Danesh Science & Technology University, Dinajpur
   Email: masud@hstu.ac.bd
   Phone: +8801737049633
   Relationship: Thesis co-supervisor and research collaborator

DECLARATION

I hereby declare that all the information provided above is true and correct to the best of my knowledge and belief.